

Project Title: AI Based Fitness & Nutrition Assistant

Problem Statement:

Many people as beginners struggle to maintain a healthy fitness routine because they do not have access to personalized workout and diet guidance that matches their goals, lifestyle, and budget. Most existing fitness platforms are designed for global users and often provide generic plans, premium-only features, or diet suggestions that are not suitable for local Bangladeshi food availability and student budgets.

As a result, beginners often become inconsistent, confused, or unmotivated during their fitness journey. A single fixed workout or diet plan is not effective for every user because fitness needs vary depending on body condition, activity level, goals, and consistency.

This project aims to solve this problem by developing an AI-assisted fitness and nutrition platform that can provide personalized workout guidance, affordable local diet suggestions, and adaptive recommendations based on user progress and fitness goals

Objectives:

The main objectives of this project are:

- Build a personalized fitness and nutrition management platform
- Generate workout and diet plans based on user profile and goals
- Track user fitness progress and consistency
- Integrate AI-based adaptive recommendations and guidance
- Suggest affordable local diet options suitable for users
- Provide a beginner-friendly and responsive user experience

Proposed Features:

- User registration and login system
- Personalized fitness profile setup
- Workout plan generation
- Budget-based diet recommendations
- Exercise library with video guidance

- Progress tracking dashboard
- AI-generated fitness recommendations
- Adaptive workout and nutrition guidance
- BMI and fitness data calculation
- Responsive user interface
- Admin management panel

Technology Stack:

Frontend:

- React.js
- Tailwind CSS

Backend:

- Node.js
- Express.js

Database:

- MongoDB Atlas

AI Integration:

- Gemini API

Deployment:

- Vercel
- Render

Expected Outcome:

The proposed system will help users maintain a more organized and personalized fitness lifestyle by providing workout and nutrition guidance based on their individual needs and goals. The AI integration will improve the user experience by analyzing fitness-related data and generating more personalized guidance, exercise explanations, and nutrition recommendations. The platform will

also focus on affordability and local relevance, making it more practical for students and beginner users in Bangladesh.